Operating Instructions

Diesel Engine with Exhaust Gas Aftertreatment
12V4000Gx5S

MS15065/04E

A Rolls-Royce solution

Engine model	kW/cyl.	Application group
12V4000G15S	124 kW/cyl.	3F, heavy duty for DCP, unrestricted, ICXN
12V4000G25S	145 kW/cyl.	3F, heavy duty for DCP, unrestricted, ICXN
12V4000G75S	137 kW/cyl.	3D, emergency standby power, IFN
12V4000G85S	160 kW/cyl.	3D, emergency standby power, IFN

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to §15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

California Proposition 65

This warning applies only to the State of California, USA, in compliance with Title 27 California Code of Regulations, Article 6, Clear and Reasonable Warnings.

⚠ **WARNING:** Breathing diesel engine exhaust exposes you to chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information go to www.P65warnings.ca.gov/diesel.

(→ www.P65warnings.ca.gov/diesel)

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1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

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2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at:http://www.mtu-solutions.com

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2.2 Important requirements for Power Generation application products

Project-specific operational environment

During integration and use of the diesel engine, all currently valid and relevant standards and guidelines on safety and specified operation must be implemented and observed. This applies to persons responsible for:
- The installation of the engine
- The design of the engine-generator set
- The installation of the engine-generator set

The relevant standards are, in particular:
- DIN EN 1679-1
- EN ISO 8528-13

Residual hazards

In the event of a defect, the diesel engine can unexpectedly have insufficient or no drive power at all. We therefore recommend a redundant design of the engine-generator set system in safety-critical applications.

Protection against hot surfaces

Combustion engines and their auxiliary components have hot surfaces (e.g. exhaust system, exhaust turbochargers, exhaust aftertreatment system, engine oil system, cooling system), which can cause injury to persons who approach or touch them (danger of burns).
- In operation
- During the cooling-down phase after operation

The plant integrator and operator must assess, based on spatial or other plant conditions, whether the danger potential usually associated with the state of technology is exceeded, e.g.:
- Particularly narrow maintenance spaces
- Particularly exposed position of hot surfaces

In such cases, the plant integrator or operator must equip the plant with the necessary and suitable safety devices to reliably exclude the possibility of personal injury, e.g.:
- Heat shields
- Insulation

The operator must ensure that the plant is not operated without these protective devices. Operating and maintenance personnel must be informed about the danger of hot surfaces and the prohibition of operation without the protective devices.

Noise

Noise protection measures that comply with statutory requirements must be implemented by the plant integrator and operator.

Chemical resistance

Only use hoses, pipes, fittings, etc. that can tolerate the fluids and lubricants in accordance with the Fluids and Lubricants Specifications. Only use options that are resistant to the fluids and lubricants in accordance with the Fluids and Lubricants Specifications.

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Electrical devices

The IP protection class of the electrical devices required by the manufacturer must be observed.

Hoses

For vibration decoupling, where possible only use hoses or rubber sleeves. Install devices in low-vibration areas and not on the engine. Hoses must be installed such that they do not chafe and are not under tension.